

Mukesh Patel School of Technology Management & Engineering
School of Technology Management and Engineering

Course Policy Document

Course Name - (Code): Elements of Project Management – (702ME0I004)

Program and Semester: B Tech programs except B Tech CSBS and B Tech CSE (DS) and Sem-V/VI	Pre-requisite Course: None					
Academic Year: 2026-27	Credit Details:	L	T	P	C	H
		3	0	0	3	3
Name of Course Faculty: Dr. Rajesh Patil Program: B. Tech (Mechanical) Contact Details: rajesh.patil@nmims.edu Office Hours: 10AM-6PM	Faculty associated with the course: - 1. Prof. Chandrkant Wani (Hyderabad Campus) 2. Prof. Mayuri Jatale-Loharkar (Shirpur Campus)					
Pre-Course Activity: Using Gen AI, explore the following: 1. Benefits of studying Project management for engineers 2. History of Projects and their executions 3. Identify how this course will help you in your career goals. 4. Identify the main timelines in the development of the project life cycle since its inception in various disciplines						
Course link: https://teams.microsoft.com/l/team/19%3AxDZjiHXQohiW0o22Sh5-UzD0ZUNrc8yUjzDLEb8S1c1%40thread.tacv2/conversations?groupId=88697ef5-5869-4a82-ac57-4a9912d92d03&tenantId=d1f14348-f1b5-4a09-ac99-7ebf213cbc81						

1. Introduction to the Course

1.1 Importance of the Course

1.1.1 Domain Relevance:

The concepts and methodologies of this course provide ground knowledge for courses like product design development which includes physical products and services. Project management remains highly relevant across a broad spectrum of domains where performance, control over schedule and efficient utilization of resources are critical factors. Its versatility and robustness continue to make it a preferred choice for Project managers and organizations tackling complex and demanding projects from variety of engineering disciplines.

1.1.2 Industry Relevance:

The case studies from Project management will cover different aspects like triple constraints of project, controlling schedule and resources, risk management, decision making abilities from different industrial sectors like manufacturing, construction, IT, Biotechnology, etc.

1.2 Objectives of the Course:

- 1.2.1** To Understand the Project Life cycle
- 1.2.2** To learn aspects of project management such as formulation, budgeting, planning, scheduling and control of projects from initiation to completion.

2. Course Outcomes (CO), Mapping with Program Outcomes (PO), and Program Specific Outcomes (PSO)

2.1 Course Outcomes

- 2.1.1 CO1:** Interpret the project feasibility and identify the role of project manager
- 2.1.2 CO2:** Relate the resources and durations using scheduling tools and techniques
- 2.1.3 CO3:** Plan the costs and risks of projects
- 2.1.4 CO4:** Apply the tools for monitoring the progress of projects

2.2 Program Outcomes (PO) the course contributed to:

- 2.2.1 PO-1:** Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems
- 2.2.2 PO-2:** Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
- 2.2.3 PO-3:** Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
- 2.2.4 PO-4:** Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

- 2.2.5 PO-5:** Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
- 2.2.6 PO-6:** The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
- 2.2.7 PO-7:** Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
- 2.2.8 PO-8:** Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
- 2.2.9 PO-9:** Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
- 2.2.10 PO-10:** Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments
- 2.2.11 PO-11:** Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

2.3 Program Specific Outcomes (PSO):

- 2.3.1 PSO-1:** Demonstrate ability to design, analyze, and optimize mechanical systems using Industry 4.0 technologies and innovations.
- 2.3.2 PSO-2:** Demonstrate competence in implementing, managing, and innovating mechanical technologies for efficient manufacturing processes

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2						3	2	2	1	1	1
CO2	2				1					2		1	
CO3						2				1		1	
CO4					2					2		1	1

Mapping Levels: 1- Low, 2-Medium, 3-High

2.4 Student Outcomes (SO) (For ABET accredited Programs):

- 2.4.1 SO-1:** Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
- 2.4.2 SO-2:** Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
- 2.4.3 SO-3:** Communicate effectively with a range of audiences.
- 2.4.4 SO-4:** Recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions.
- 2.4.5 SO-5:** Function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
- 2.4.6 SO-6:** Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
- 2.4.7 SO-7:** Acquire and apply new knowledge as needed, using appropriate learning strategies.

3. Teaching-learning methodology**3.1 Instruction Plan**

Lecture No.	Topic	Teaching Method *	Blooms Level	Resources	COs mapped	Assessment and Evaluation
Unit 1: Introduction to project management:						
1	Where do the projects come from?	PPT, Whiteboard	Understand	T1 pp. 1 - 2	CO1	Class Test-1 and TEE
2	What is Project Management		Understand	T1 pp. 1 - 2	CO1	

3	Classification of project management in different environments		Understand	R1 pp. 5	CO1	
4	Definition of System, System Structure, System Approach		Understand	T1 pp. 9	CO1	
5	Project Feasibility, Project Life Cycle		Understand	T1 pp. 4-22	CO1	
6	Phases of System Development Cycle		Understand	T1 pp. 22	CO1	
7	Case Study on Software Product Development		Apply	--	CO1	
8	Case Study on Need of Project Management		Apply	--	CO1	
Unit 2: Project Organization						
9	Project Integration	PPT, Whiteboard	Understand	R1 pp. 127	CO1	Class Test-1 and TEE
10	Project organization, Types of structures		Understand	R1 pp. 127	CO1	
11	Roles of Project managers and team		Understand	T1- pp.79 R1- pp 155	CO1	
12	Integration in large scale projects.		Understand	R1 pp. 127	CO1	
13	Case study on Need of Org. Structure/Project Manager		Apply	--	CO1	
Unit 3: Project Planning and Scheduling						
14	What is Project Planning and Project Scheduling	PPT, Whiteboard	Understand	T1 pp. 171	CO2	Class Test-2 and TEE
15	Steps in Project Planning		Understand	T1 pp. 171	CO2	
16	Need of Project Planning		Understand	T1 pp. 171	CO2	
17	Project Master Plan, Elements of Master Project Plan		Create	T1 pp. 183	CO2	
18	Work Breakdown Structure		Create	T1 pp. 183	CO2	
19	Other Tools of Project Planning, Work Packages		Create	T1 pp. 183	CO2	
20	Case Study on Work Break Down structure		Create	T1 pp. 183	CO2	
21	Responsibility Matrix, Gantt Chart		Create	T1 pp. 192	CO2	

22	Network Diagram		Evaluate	T1 pp. 186	CO2	
23	Network Diagram		Evaluate	T1 pp. 186	CO2	
24	Critical Path, Activity-on-Arrow		Evaluate	T1 pp. 278-279	CO2	
25	CPM & PERT for Scheduling and Resource allocation		Evaluate	T1 pp. 278-279	CO2	
Unit 4: Cost Estimating and Budgeting						
26	Basics of cost estimating and its elements	PPT, Whiteboard	Understand	T1 pp. 225	CO3	Class Test-2 and TEE
27	Budgeting of the project		Understand	T1 pp. 229-230	CO3	
28	Project cost accounting		Understand	T1 pp. 225	CO3	
29	Management information systems		Understand	T1 pp. 38	CO3	
30	Cost schedules and forecasts		Apply	T1 pp. 234-236	CO3	
31	Case Study on Cost estimation & Budgeting		Apply	--	CO3	
Unit 5: Project Risk Management						
32	Basic concept of risk, Risk Concepts and Identification	PPT, Whiteboard	Understand	T1 pp. 22, 245, 246	CO3	TEE
33	Risk identification, risk response planning		Understand	T1 pp.178-179	CO3	
34	Risk management methods		Understand	T1 174	CO3	
35	Information monitoring, Control process		Understand	T1 387, 402-403	CO4	
36	Information monitoring, Control process		Understand	T1 387, 402-403	CO4	
37	Cost accounting systems for project control		Understand	T1 PP.225, 374-376, 387	CO4	
38	Cost accounting systems for project control		Understand	T1 PP.225, 374-376, 387	CO4	
39	Performance analysis		Understand	T1 293-294	CO4	
40	Performance analysis		Understand	T1	CO4	

				293-294		
Unit 6: Evaluation and Termination						
41	What is Evaluation and termination	PPT, Whiteboard	Understand	T1 pp. 449, 455-456	CO4	TEE
42	Project reviews and reporting		Understand	T1 pp. 455-472	CO4	
43	Project reviews and reporting		Understand	T1 pp. 455-472	CO4	
44	Closing the contract.		Understand	T1 pp. 455-472	CO3	
45	Case study on Closing the contract.		Understand	--	CO3	

4. Assessment and Evaluation Scheme

	Internal Continuous Assessment (ICA) (50 Marks)				
	Class Tests (20 Marks)		Term-work (30 Marks)		
Assessment Component	Class Test-1	Class Test-2	Case Study Presentations	Assignments (In-class and Home)	Class Participation
Marks	10	10	15	10	5

4.1 Internal Continuous Assessment (ICA) – 50 marks

Class Test-1 conducted in the 6th week. It will be for 10 marks (1-hour duration). Class Test-2 conducted in the 11th week. It will be for 10 marks (1-hour duration). In-class and home assignments given will carry a weightage of 10 marks. Evaluation based on timely submission. 50% marks would be deducted for late submissions. (It is important that all assignments submissions are done before the deadline given). Class participation will carry a weightage of 5 marks. Case study presentations will carry a weightage of 15 marks.

Rubrics for case study preparation and presentation:

Assessment Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)	Marks
1. Understanding of the Case and Problem Identification	Clearly explains the case background and accurately identifies all major project management issues with excellent understanding.	Explains the case well and identifies most important issues.	Basic understanding with only some issues identified.	Limited understanding; fails to identify key issues.	3
2. Application of Project Management Concepts	Correctly applies appropriate project management concepts, tools, and techniques (scope, time, cost, risk, stakeholders, communication, etc.) with logical justification.	Applies relevant concepts with minor errors or omissions.	Applies only a few concepts with limited justification.	Concepts are incorrectly applied or largely missing.	4
3. Analysis, Recommendations, and Conclusions	Performs critical analysis, evaluates alternatives, provides practical recommendations, and draws well-supported conclusions.	Good analysis with reasonable recommendations and conclusions.	Basic analysis with generic recommendations.	Little or no analysis; recommendations are unclear or unsupported.	3
4. Presentation Skills and Communication	Presentation is well-organized, confident, within time, visually appealing, and answers questions effectively.	Good organization and communication with minor presentation issues.	Adequate presentation but lacks confidence or clarity.	Poor organization, ineffective communication, or unable to answer questions.	3
5. Teamwork and Participation	All members contribute equally, demonstrate excellent coordination and collaboration.	Most members contribute effectively.	Unequal participation; coordination needs improvement.	One or two members dominate; poor teamwork.	2
Total					15 Marks

4.2 Term End Examination (TEE) – (100 marks scaled down to 50)

TEE conducted at the end of the semester will be for 100 marks (3-hour duration). Marks obtained will be scaled down to 50. There will be 7 questions (each question will be 20 marks). Q1 will be compulsory and any 4 from the remaining to be solved.

4.3 Course Passing Criteria

4.3.1 ICA (50 marks) – No minimum marks

4.3.2 TEE (100 marks scaled to 50) – 40% required for passing

4.3.3 (ICA + TEE) (100 marks) - 50% required for passing

4.4 Assessments and Mapping to Course Outcomes

	Internal Continuous Assessment (ICA)				Term End Examination (TEE)
Course Outcomes	CT-1	CT-2	Case Study Presentations	Assignments	TEE
CO-1	Y		Y	Y	Y
CO-2		Y	Y	Y	Y
CO-3		Y	Y	Y	Y
CO-4			Y	Y	Y

5. Laboratory details

This course does not have laboratory.

6. Tutorial Plan

This course does not have any tutorial.

7. Course Material

References and Lab Manuals would be uploaded on LMS by faculty every week.

8. GenAI Usage

8.1 Pre-class Activity:

- AI-Generated Reading Material
- Interactive AI Simulation
- Case Study Analysis

8.2 In-Class Activity:

- Introduction and Team Formation
- Problem-Solving and Decision-Making
- Debrief and Discussion

8.3 Assignments:

GenAI is not to be used for doing Assignments given by faculty.

8.4 Case Study Presentations:

GenAI is permitted to be used. Students need to cite the references used.

9. Academic Integrity Statement

Original work expected from students for all of the **assigned assessment work**. Copying in any form not acceptable and will invite strict disciplinary action. Evaluation of corresponding component will be affected proportionately in such cases. Plagiarism detection software will be used to check plagiarism wherever applicable. Academic integrity is expected from students in all components of course assessment.

*** - Only Teaching Method in the Instruction Plan for the course may vary for different faculty teaching the course, rest of the Course Policy Document will not change.**